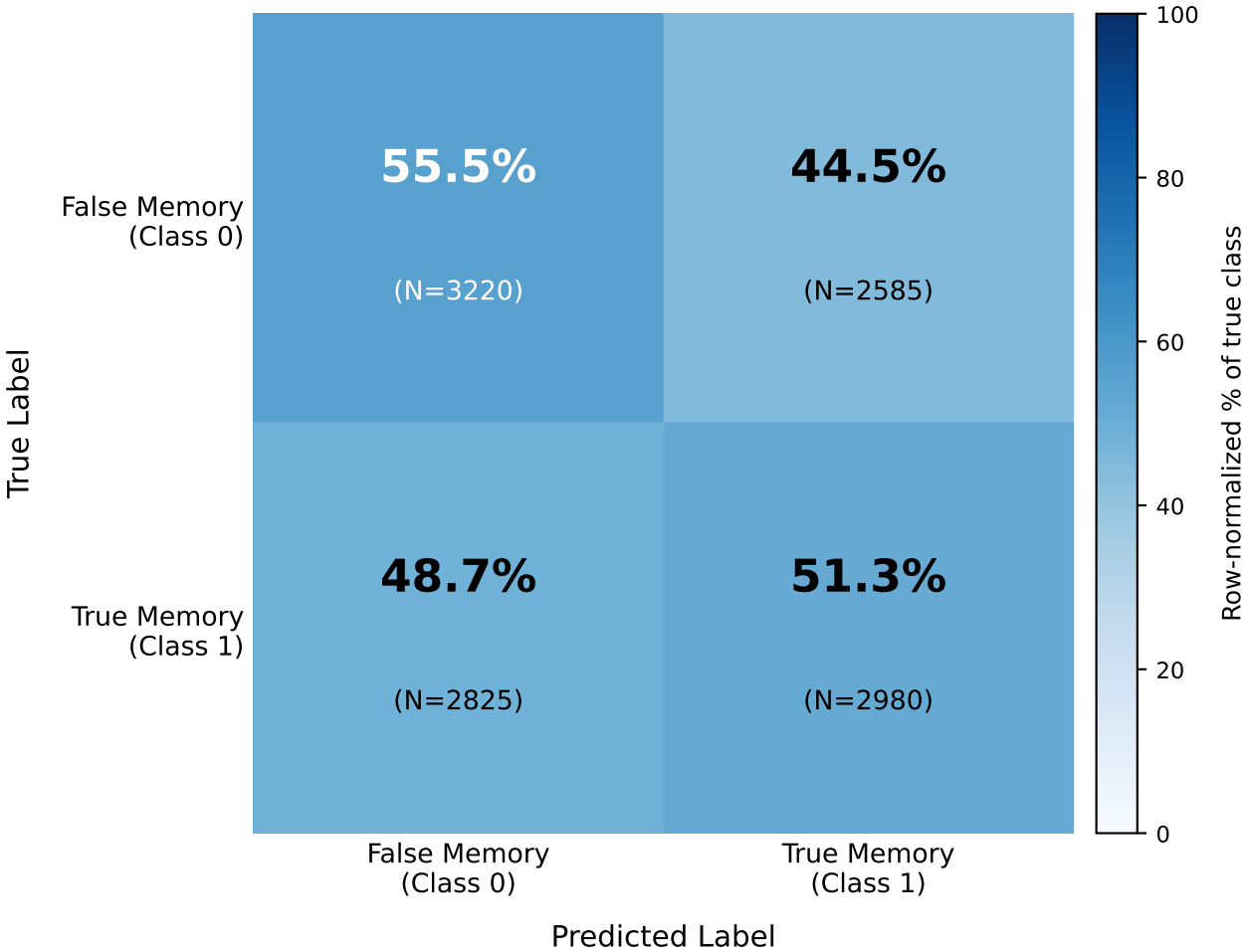


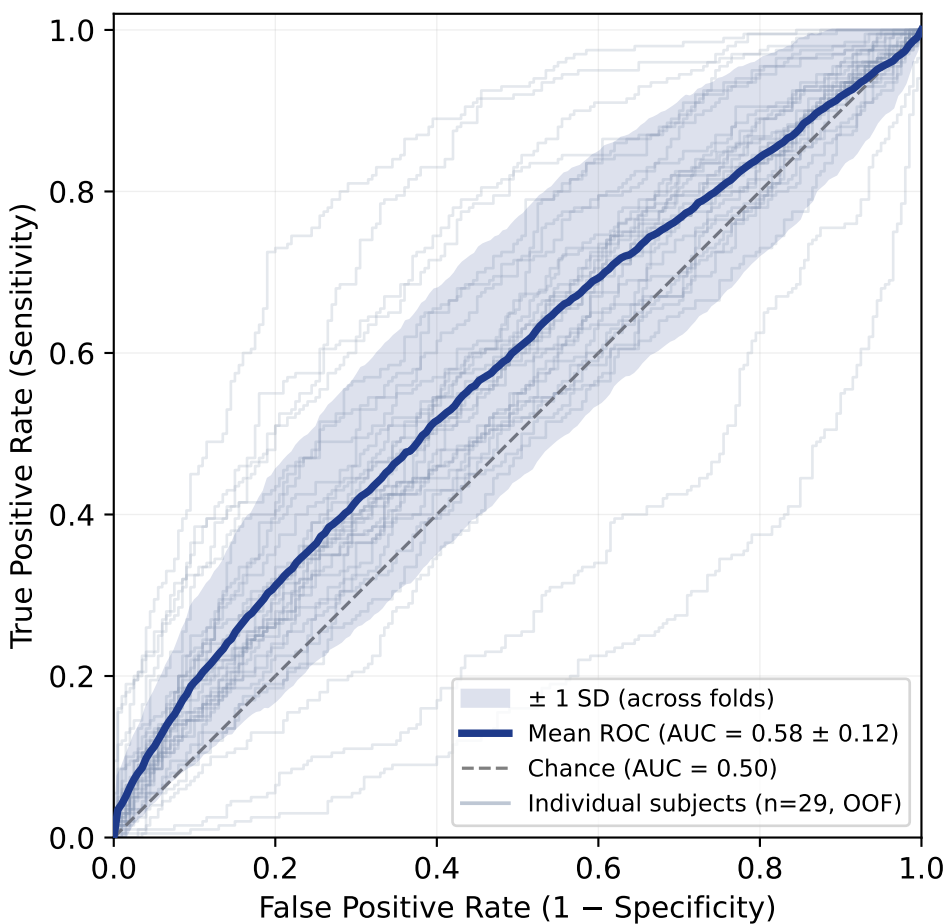
Figure 5: Overall Classifier Diagnostics Across 29 Subjects (Aggregated Confusion Matrix & ROC-AUC)

Pipeline: EA-aligned tangent features → fold-wise StandardScaler → Shrinkage-LDA (Ledoit-Wolf) | feature path: precomputed_tangent_features

**A. Aggregated Confusion Matrix
(LOSO, Out-of-Fold, Row-Normalized)**



**B. ROC Curve Analysis
(Leave-One-Subject-Out, N=29)**



All confusion-matrix cells and ROC curves reflect strictly out-of-fold (OOF) predictions pooled across 29-fold Leave-One-Subject-Out cross-validation — no held-out subject's trials were seen during model fitting at any stage.